Fig. 1 (PDF p.3): multibranch-to-decoder layout

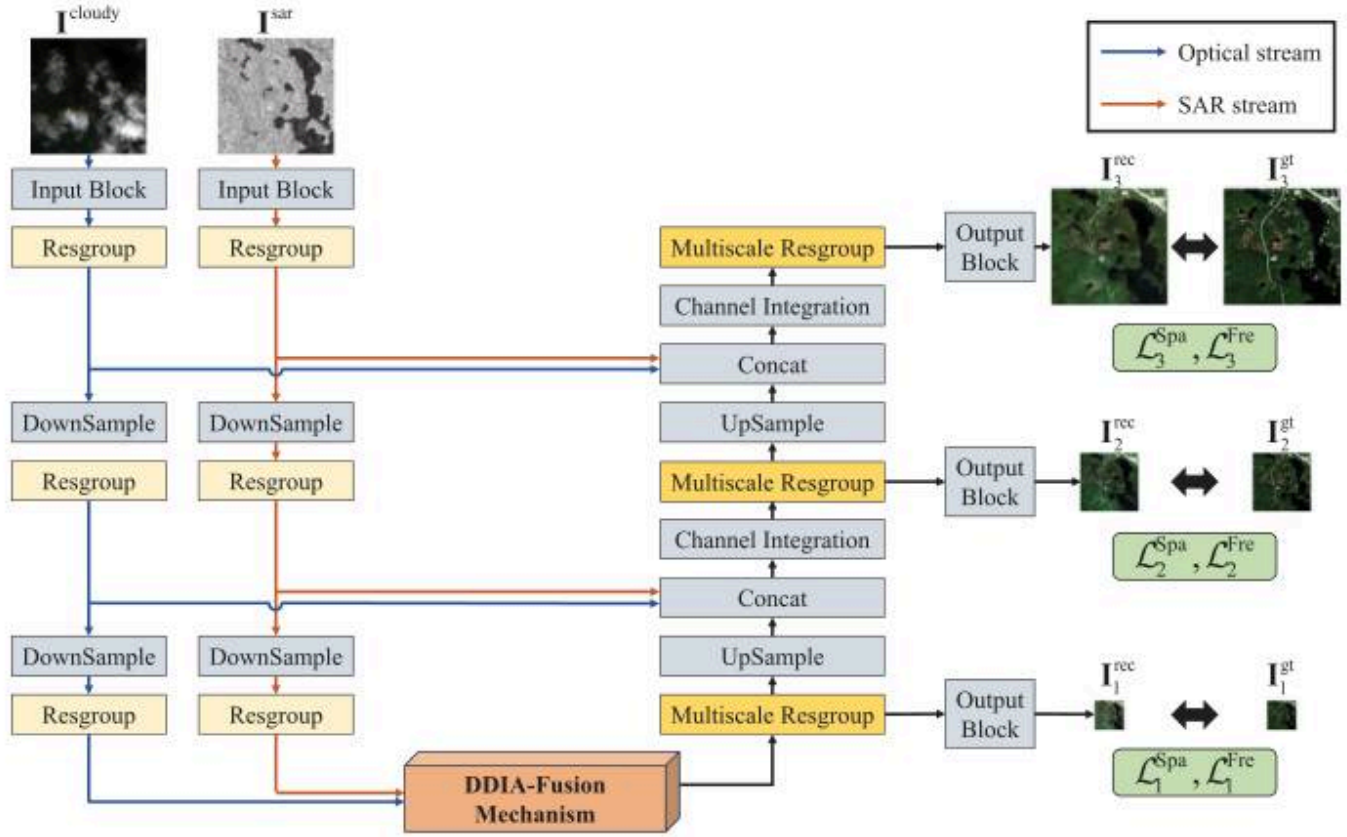


Fig. 1. Architecture of the proposed DDIA-CFR network. The overall pipeline is a W-shaped asymmetric encoder–decoder network including a two-branch encoder, a multiscale and multi-output decoder, skip connections, and a dual-domain interactive-attention-based feature fusion bottleneck.

3. To enhance the effectiveness of attention interaction in cross-modality fusion, features in the frequency domain are also concerned and used to supplement features that are not salient into the bottleneck structure to facilitate effective and rational cross-modality feature fusion. For the specially designed bottleneck, the dual-domain interactive-attention feature fusion mechanism is used to

Study: two input lanes, consistent stages, colored flows; real input/output; preserve structure-dependent geometry.

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Fig. 3 (PDF p.5): module insets and multi-type links

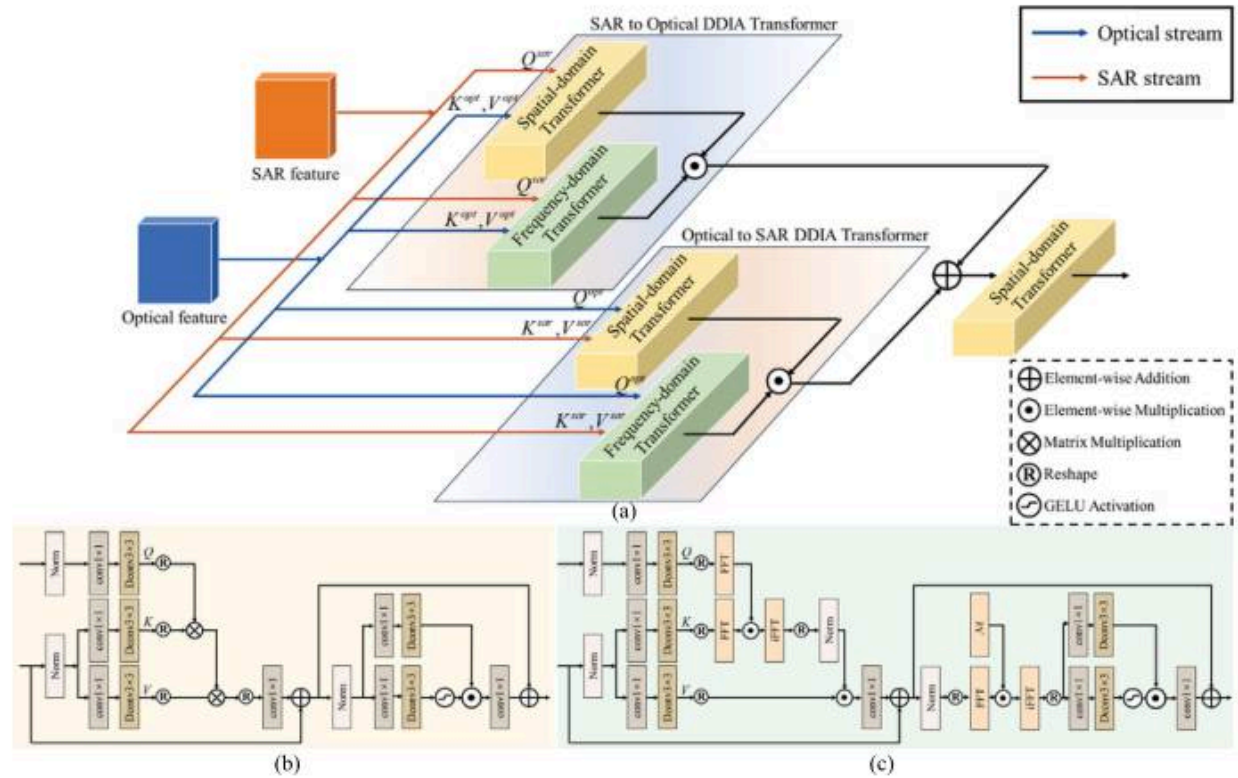


Fig. 3. Architectures of Dual-Domain Interactive-Attention (DDIA) based feature fusion mechanism. (a) Interactive-Attention based feature fusion strategy. (b) Spatial-domain Transformer block. (c) Frequency-domain Transformer block.

2.2.2. Optical to SAR interactive-attention transformer group

Contrary to the S-ODDIA transformer, the O-SDDIA transformer group is designed to transfer information from the optical image to SAR features. Given that the *query* Q^{opt}, *key* K^{sar}, and *value* V^{sar} tokens are calculated in the same way as S-ODDIA transformer through the projection matrices W_i^P, the *query* in O-SDDIA transformer group is

2.3. Dual-domain transformer block

The significant information loss caused by thick clouds hampers the ability of overall image perception from the spatial domain. As a consequence, the reconstruction results may suffer from potential artifacts or semantic ambiguity. Extensive studies have revealed that

Study: 2.5D feature blocks, separate subfigures, a precise operation legend, concise functional labels.

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